**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 315 · Networks, security and compression practice

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does DNS resolve?

Q2. What is a router's role?

Q3. What is lossy compression?

Q4. $11010011_2$ AND $00001111_2$ as byte and hex?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

DNS — Maps names to IP addresses.

TLS — Authenticated secure transport.

RLE — Run-length encoding.

digest — Output of hash function.

## New learning

Understand — Distinguish DNS addressing, IP routing, TCP transport and TLS security: each serves a different purpose.

Apply — RLE is lossless and suitable for runs; dictionary coding uses references. Lossy methods can discard detail.

Evaluate — Hashing produces an integrity digest; symmetric and asymmetric encryption protect confidentiality using keys.

## Worked example

Encode AAABBCC with run-length encoding.

1. Predict first — Count consecutive symbols, preserving the order of groups.

2. Explain and check — Explain the reason for the result.

# We Do · Practise with guidance

Explain the limitation of a checksum displayed on the same compromised download website.

```
Attacker could alter both file and checksum.
```

Predict first

A matching hash is evidence of integrity only when its expected value comes from a trustworthy channel.

Explain and check

Explain the reason for the result.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Describe DNS, router and TLS roles when accessing HTTPS website. [6]

### Question 2

RLE-encode AAAAABBBCCDD and explain how decoding works. [6]

### Question 3

Evaluate transmitting official exam results as a lossy image versus an authenticated, encrypted, losslessly stored file. [6]

# Exit Ticket · Answer before checking

Hashes can be decrypted using the public key to recover the input.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Domain name to an IP address.

Q2. Forwards IP packets between networks.

Q3. Discards information irreversibly.

Q4. $00000011_2 = 03_{16}$.

### You Do pages · Independent questions

Q1. DNS maps name/IP [2], router forwards between networks [2], TLS protects/authenticates channel [2].

Q2. (A,5) [1],(B,3) [1],(C,2) [1],(D,2) [1]; repeat each character count times [2].

Q3. Exact data needed [2], lossless preserves digits [1], TLS/confidentiality [1], authenticity/integrity via trusted mechanism [1], access permissions [1].

### Exit Ticket · Model answer

False. A hash is not reversible encryption.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**